A Comparative Study on Transfer Learning Strategies for Fine-Grained Waste Image Classification

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Abstract

With the acceleration of urbanization, waste sorting has become a cornerstone of sustainable environmental management. However, due to the large number of waste categories and the high visual similarity between different materials, automated waste sorting faces substantial challenges, particularly in fine-grained classification under class imbalance. This work conducts an empirical study of three deep transfer learning strategies built on the VGG16 architecture, evaluated on the RealWaste dataset, which contains 4,752 images across 9 categories: (1) a frozen backbone with cost-sensitive class weights, (2) layer-specific fine-tuning with cost-sensitive class weights, and (3) layer-specific fine-tuning with data-level oversampling. We incorporate stratified sampling, geometric and photometric data augmentation, and two complementary strategies for handling the long-tailed distribution. Experimental results demonstrate that fine-tuning offers significant advantages in capturing fine-grained, domain-specific features. Compared to the frozen baseline, the fine-tuned model with cost-sensitive weights achieves a 9.3 percentage-point improvement in overall accuracy and over 10 points in both weighted and macro F1 on the test set. The oversampling experiment further reveals a meaningful trade-off between data-level and algorithm-level imbalance correction. This report details the model design, full evaluation, confusion-matrix analysis, and an ablation study across all three strategies.

Introduction

Automated waste-sorting systems are vital for modern smart cities, as traditional manual sorting is inefficient and hazardous. While deep learning offers immense potential, real-world applications face three primary machine-vision challenges: (1) complex physical deformations and background clutter; (2) fine-grained classification difficulty, where different materials appear visually similar while identical materials vary drastically in shape; and (3) severe data scarcity characterized by a long-tailed distribution, in which majority classes heavily overshadow scarce categories.

To overcome the overfitting risks of training deep CNNs from scratch on limited data, transfer learning leverages pre-trained ImageNet weights to provide robust low-level visual representations.

This project utilizes the classic VGG16 backbone for its clear hierarchical architecture, which enables precise control over convolutional layers at varying depths and facilitates the study of feature-extraction differences across blocks.

Our core motivation is to systematically quantify how varying transfer-learning depths and imbalance-handling strategies affect fine-grained, imbalanced waste classification. We establish a complete experimental pipeline on the RealWaste dataset, contrasting a baseline frozen-backbone strategy against a layer-specific fine-tuning approach, and further introducing a data-level oversampling ablation. To explicitly mitigate the dataset's long-tailed distribution, we integrate stratified sampling, data augmentation, and two complementary techniques: inverse-frequency class weights (algorithm-level) and `RandomOverSampler` (data-level).

Preliminary results confirm our hypothesis: fine-tuning the high-level convolutional block significantly enhances the capture of fine-grained discriminative features. Compared to the frozen baseline, the fine-tuned model improves overall accuracy by approximately 9.3 percentage points on the independent test set and achieves substantial gains in recall across multiple minority classes.

Literature Review

Convolutional Neural Networks (CNNs) are the standard tool for image classification. VGGNet [1] demonstrated that stacking 3×3 kernels enables deeper architectures and yields strong visual representations, making it a widely used backbone for transfer learning. Pan and Yang [2] surveyed transfer learning and emphasized that differences between source and target domains and tasks, such as feature spaces and data distributions, largely determine which transfer setting and methods are appropriate. In waste classification, Yang and Thung [3] explored learning-based baselines on the TrashNet dataset, including a CNN-style model, and discussed practical techniques such as preprocessing and augmentation under data constraints.

To further understand feature transferability, Yosinski et al. [6] showed that early layers in deep neural networks learn general, Gabor-like filters applicable to almost any image dataset, while deeper layers become highly task-specific. This foundational insight motivates our methodology of freezing the lower layers of VGG16 and unfreezing only the highest convolutional block to adapt the network to domain-specific waste textures. Recent applications in waste management have likewise demonstrated the efficacy of deep transfer learning. Adedeji and Wang [7] showed that pre-trained architectures (such as ResNet and VGG) significantly reduce convergence time and improve robustness on small-scale waste datasets compared to training networks from scratch.

However, real-world waste data inevitably exhibits long-tailed distributions [4], in which majority classes overshadow scarce categories. To combat the inductive bias caused by class imbalance, techniques that focus on the minority, such as the principles behind Focal Loss introduced by Lin et al. [8], highlight the necessity of strongly penalizing easy majority-class predictions, a concept we implement via inverse-frequency class weights. Furthermore, Costa et al. [9] emphasize that for fine-grained waste classification under data constraints, robust data augmentation and careful regularization of the classification head (e.g., Dropout) are critical to prevent overfitting. Building on these insights, and aligning with Kornblith et al.'s [5] finding that fine-tuning pre-trained features yields superior adaptation, this project systematically evaluates how fine-tuning depth and imbalance-handling strategy affect generalization on the highly imbalanced RealWaste dataset.

Technical Plan

The core objective of this project is to systematically evaluate transfer-learning strategies of varying depths in a real-world scenario with limited and highly imbalanced data. To this end, we designed a complete deep-learning pipeline comprising data preprocessing, dynamic augmentation, a customized classification head, and three sets of comparative fine-tuning and imbalance-handling strategies. All models were built and trained using the TensorFlow framework.

Dataset and Stratified Splitting

The RealWaste dataset was used, containing 4,752 high-resolution images of household waste across 9 categories. Data analysis revealed a typical long-tailed distribution with an imbalance ratio of 2.9×. The largest classes, Plastic (19.4% of training samples) and Metal (16.6%), were dominant, while the most underrepresented class, Textile Trash, accounted for only 6.7% of the training set.

To ensure unbiased model evaluation and prevent further imbalance during data splitting, we adopted stratified sampling with a fixed random seed (`SEED=42`). The dataset was divided into training, validation, and independent test sets at 64%, 16%, and 20% respectively, yielding 3,040/761/951 images. This strategy ensures that the class proportions in each subset are consistent with the global data distribution, providing a reliable foundation for the subsequent metric calculations.

Preprocessing and Data Augmentation Pipeline

Since waste in real-world environments is often compressed, damaged, or placed at arbitrary angles, we built a robust preprocessing and augmentation module.

Normalization. All input images are first resized to 224×224 to fit the standard receptive field of VGG16. We then use Keras' native `vgg16_preprocess_input` function to convert pixel values to floating-point numbers and apply ImageNet mean normalization, maximizing compatibility with the pre-trained weights.

Geometric and Photometric Augmentation. During training, we use `keras.Sequential` to construct real-time data augmentation layers. Specific operators include random flip, zoom, translation, and contrast adjustment. These transformations significantly expand the effective sample distribution, forcing the model to learn intrinsic object textures rather than memorizing specific positions or poses, and thereby mitigating overfitting.

Model Architecture and Layer-Specific Fine-Tuning

Our experiments use VGG16 pre-trained on ImageNet as the backbone. We remove its original high-dimensional fully connected layers and design a classification head adapted to the 9-class task.

Classification Head. The output of the backbone is first reduced by a global average pooling layer, which substantially lowers the parameter count compared with the traditional flattening operation. This is connected, in sequence, to: a batch normalization layer, a dense layer with 512 units (ReLU activation, with weight regularization), a second batch normalization layer, a dropout layer with rate 0.2, and finally a softmax output layer with 9 units.

Controlled Experiments.

1. **Frozen VGG16 + Cost-Sensitive Weights.** All 13 convolutional layers of the VGG16 backbone are frozen, turning it into a static feature extractor. Only the custom classification head is updated during training. Inverse-frequency class weights are applied during loss computation.

2. **Fine-Tuned VGG16 + Cost-Sensitive Weights.** A local unfreezing strategy is employed: the `trainable` attribute of the last convolutional block of VGG16 (`block5_conv1`, `block5_conv2`, `block5_conv3`) is set to `True`. The rationale is that lower-level convolutions retain the ability to extract general edges and colors, while higher-level filters are allowed to update so as to capture fine-grained visual features specific to RealWaste (e.g., the reflectivity of glass bottles or the wrinkled texture of cardboard). Inverse-frequency class weights are kept identical to those in Experiment 1.

3. **Fine-Tuned VGG16 + Oversampling (Ablation).** This experiment uses the same fine-tuned block5 backbone as Experiment 2 but replaces the algorithm-level class weighting with a data-level balancing strategy. `RandomOverSampler` from the `imbalanced-learn` library is applied to the training file-path array, duplicating minority-class samples until all 9 classes match the count of the majority class (Plastic: 589). This expands the effective training set from 3,040 to 5,301 images. Critically, `class_weight` is set to `None` to avoid double-penalizing minority classes at both the data and loss levels.

Training Configuration and Class Weight Penalization

Model training consistently used the Adam optimizer with categorical cross-entropy loss. Due to the large size of our custom classifier head, the initial learning rate was set to 10^{-3}, decaying to 5×10^{-4} in the later stages of training to ensure stable convergence. The batch size was 64, and asynchronous data prefetching was enabled via `tf.data.AUTOTUNE` to optimize GPU/CPU I/O throughput.

For Experiments 1 and 2, in addition to data augmentation, we introduced inverse-frequency class weights during loss computation to address the long-tailed distribution. Based on the true frequencies in the training set, we down-weighted the dominant Plastic class (weight = 0.573) and Metal class (weight = 0.669), while up-weighting the most underrepresented Textile Trash class (weight = 1.664). This mechanism encourages the model to focus more on minority classes during optimization, improving the balance of recall across categories. Experiment 3 instead relies solely on the oversampling strategy described above and does not apply class weights.

Experimental Results

We evaluated all three models on the independent test set of 951 images. The results confirm our central hypothesis: appropriately unfreezing high-level convolutional layers significantly enhances feature extraction for fine-grained, imbalanced classification, and the choice of imbalance-handling strategy introduces a meaningful performance trade-off.

Overall Performance Comparison

With identical classification heads and hyperparameters across all three experiments, the results are summarized in Table 1.

Table 1: Overall performance comparison on the test set (951 images).

Experiment	**Accuracy**	**Weighted F1**	**Macro F1**	**AUC (OVR-W)**
Exp 1: Frozen + Cost-Sensitive	60.6%	59.7%	59.2%	92.2%
Exp 2: Fine-Tune + Cost-Sensitive	**69.9**%	**69.7**%	**69.5**%	**94.6**%
Exp 3: Fine-Tune + Oversampling	67.0%	66.5%	67.7%	94.5%
Improvement (Exp 1 → Exp 2)	+9.3 pp	+10.0 pp	+10.3 pp	+2.4 pp

Experiment 2 achieves the highest scores on every overall metric. The simultaneous gains of more than 10 percentage points in both weighted and macro F1 indicate that fine-tuning combined with cost-sensitive weighting effectively adapts the ImageNet pre-trained representations to the RealWaste domain, producing balanced cross-class improvements rather than overfitting to majority classes.

Experiment 3 (oversampling) yields an intermediate result: its accuracy (67.0%) and weighted F1 (66.5%) fall between those of Experiments 1 and 2, but its macro F1 (67.7%) exceeds the frozen baseline by 8.5 percentage points. Notably, Experiment 3 attains the highest per-class F1 among all three experiments for Cardboard (0.789) and a score nearly tied with Experiment 2 for Vegetation (0.822 vs. 0.828), suggesting that data-level oversampling can selectively benefit certain minority classes even when overall performance trails the cost-sensitive fine-tuning strategy.

Per-Class Performance Observations

Table 2 reports per-class F1 scores for all three experiments, enabling a direct ablation comparison.

Several patterns are noteworthy. First, fine-tuning (Experiment 2) substantially improves F1 across most classes, with the largest gains on Glass (+18.6 pp), Metal (+24.3 pp), and Vegetation (+25.2 pp). Second, Miscellaneous Trash remains the hardest class across all three experiments (F1 ≤ 0.53), reflecting its inherent visual heterogeneity. Third, oversampling (Experiment 3) hurts classes for which the majority class Plastic is a common confusion target, likely because duplicated minority-class samples introduce noise near the Plastic decision boundary.

Error Analysis

Despite the 9.3 percentage-point accuracy gain from Experiment 1 to Experiment 2, the absolute accuracy of 69.9% highlights the inherent difficulty of this fine-grained task. An analysis of the

Table 2: Per-class F1 score comparison across all three experiments.

Class	Exp 1 (Frozen+CW)	Exp 2 (FT+CW)	Exp 3 (FT+OS)
1-Cardboard	0.677	0.672	**0.789**
2-Food Organics	0.607	**0.721**	0.704
3-Glass	0.582	**0.768**	0.692
4-Metal	0.492	**0.735**	0.582
5-Miscellaneous Trash	0.449	**0.526**	0.509
6-Paper	0.690	0.653	**0.713**
7-Plastic	**0.700**	**0.700**	0.636
8-Textile Trash	0.561	**0.653**	0.646
9-Vegetation	0.576	**0.828**	0.822

test-set confusion matrices (Figure 1) reveals the dominant confusion pairs in each experiment.

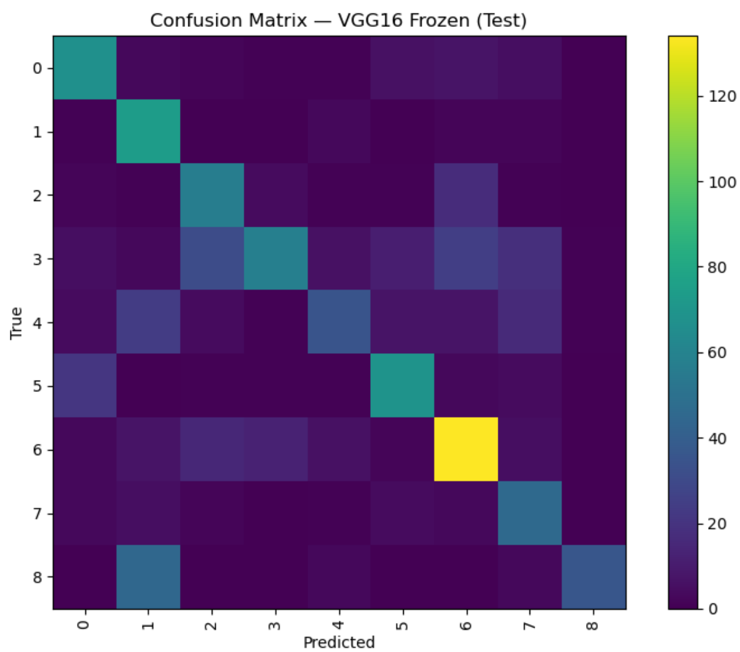

Figure 1: Confusion matrix comparison on the test set.

For **Experiment 1 (Frozen)**, the two largest confusion pairs are:

- **Vegetation → Food Organics (45 errors).** The frozen ImageNet features fail to distinguish green organic material (plant stems, leaves) from food waste, since both share similar color and texture statistics at the low-level feature representation level.
- **Metal → Glass (31 errors) and Metal → Plastic (25 errors).** Metallic objects (cans, containers) are frequently confused with translucent or reflective materials, as the frozen backbone cannot adapt its reflectance-detection filters to the RealWaste domain.

For **Experiment 2 (Fine-Tuned with Cost-Sensitive Weights)**, fine-tuning block5 substantially resolves the Vegetation/Food Organics confusion but shifts the primary errors to:

- **Plastic → Metal (27 errors).** Rigid plastic containers share shape and surface gloss with metal cans, a boundary that remains challenging even for domain-adapted features.
- **Paper → Cardboard (26 errors).** Crumpled or torn paper samples continue to be confused with cardboard, owing to high intra-class variance and strong similarities in color and base material.

For **Experiment 3 (Oversampling)**, the dominant error is **Plastic → Miscellaneous Trash (35 errors)**, suggesting that oversampling minority classes increases false-negative pressure on the Plastic majority class, redistributing rather than eliminating confusion.

Conclusion

This project conducted a systematic ablation study of three transfer-learning strategies for fine-grained waste image classification on the RealWaste dataset, a 9-class, 4,752-image benchmark with a characteristic long-tailed distribution (imbalance ratio 2.9×). The three experiments share an identical VGG16 backbone and classification head, differing only in the combination of fine-tuning depth and imbalance-handling strategy, which enables a clean and controlled comparison.

The central finding is that **domain adaptation through selective layer unfreezing is the single most impactful lever in this setting.** Fine-tuning the final convolutional block (block5) while applying inverse-frequency class weights (Experiment 2) yields an accuracy of 69.9% and a macro F1 of 69.5%, improving over the fully frozen baseline (Experiment 1) by 9.3 percentage points in accuracy and 10.3 in macro F1. This confirms the theoretical expectation from Yosinski et al. [6] that high-level VGG16 filters encode ImageNet-specific features that must be adapted to capture domain-specific textures of waste materials, such as metallic reflectance, glass transparency, and cardboard fibre patterns.

The ablation between algorithm-level and data-level imbalance correction (Experiments 2 vs. 3) reveals a more nuanced trade-off. Replacing cost-sensitive class weights with `RandomOverSampler` (Experiment 3) reduces overall accuracy by 2.9 percentage points and weighted F1 by 3.1 points, yet yields the highest per-class F1 for Cardboard (0.789 vs. 0.672) and a near-tied score on Vegetation (0.822 vs. 0.828). This asymmetry suggests that oversampling redistributes model attention non-uniformly: it benefits classes whose minority duplicates are visually coherent and distinguishable, while introducing boundary noise near the Plastic majority class. This is evidenced by the sharp drop in Plastic F1 from 0.700 to 0.636 and the emergence of Plastic → Miscellaneous Trash as the single largest confusion pair (35 errors). The two strategies are therefore complementary rather than interchangeable, and neither dominates across all nine classes.

Despite an AUC exceeding 94% in both fine-tuned models, indicating strong probabilistic ranking ability, absolute accuracy remains capped at approximately 70%. The Miscellaneous Trash class constitutes the persistent bottleneck (F1 ≤ 0.53 across all experiments), attributable to its definition as a heterogeneous catch-all category rather than a visually coherent class. Similarly, the Paper/Cardboard and Plastic/Metal confusion pairs survive all three training regimes, pointing to irreducible inter-class visual overlap that architecture-level and sampling-level interventions alone cannot resolve.

In summary, this study demonstrates that for a constrained, imbalanced waste-classification task: (1) selective fine-tuning of high-level convolutional features is necessary and largely sufficient to achieve the largest performance gain; (2) cost-sensitive class weighting provides a more globally stable imbalance correction than random oversampling on this dataset; and (3) the ceiling of roughly 70% accuracy under the current single-backbone, single-head design calls for more fundamental architectural and data-curation advances in future work.

Future Work

The experimental findings surface several concrete, empirically motivated directions for further investigation.

- **Hybrid Imbalance Strategy.** Experiments 2 and 3 demonstrate that cost-sensitive weighting and random oversampling have complementary class-level strengths: Experiment 2 dominates on Glass, Metal, and Food Organics, whereas Experiment 3 dominates on Cardboard and Paper. A natural next step is a *class-selective* hybrid that applies oversampling only to the subset of minority classes where it demonstrably helps (those with visually coherent intra-class structure), while retaining cost-sensitive weights for classes such as Plastic whose decision boundary is destabilized by duplicated samples. Alternatively, Focal Loss [8] could replace static inverse-frequency weights, dynamically down-weighting easy majority-class examples at each training step rather than relying on pre-computed frequency statistics.
- **Deeper and Progressive Fine-Tuning.** The current design unfreezes only block5 (3 layers). The persistent Metal $\rightarrow$ Glass and Plastic $\rightarrow$ Metal confusions suggest that block4 filters, which encode mid-level shape and surface-normal statistics, may also require domain adaptation for metallic and transparent materials. A progressive unfreeze schedule (block5 first, then block4, with learning-rate warm-up at each stage) would allow systematic measurement of the accuracy-versus-overfitting trade-off as the unfrozen depth increases, providing a more complete characterization of the fine-tuning landscape for this dataset.
- **Explainability via Grad-CAM.** All three experiments exhibit a common failure mode: high-confidence misclassifications in which the model's predicted probability exceeds 0.8 yet the true label is incorrect. Gradient-weighted Class Activation Mapping (Grad-CAM) [10], applied to these high-confidence errors, would reveal whether the model is attending to discriminative object regions (can ridges, bottle necks) or to spurious background cues (collection bins, ground textures). This analysis is particularly critical for the Paper $\rightarrow$ Cardboard pair, where the decision boundary may hinge on low-level surface texture that is easily corrupted by image compression artifacts in the RealWaste dataset.

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